

How to Measure RSI

Nine data asks for frontier AI companies

Cheryl Wu*, Arjun Ramani*, Basil Halperin*

Lukas Althoff, Brian Jabarian, Andrew Koh, Phil Trammell

*Lead authors

CAPABILITIES RSI would accelerate capability growth, and the acceleration would show up in internal models before released ones.

- 1 Internal-model benchmark scores.** Results on a diverse set of benchmarks for the latest internally deployed models, including unreleased ones, aggregated into Epoch's Capabilities Index (ECI) or an internal equivalent. [Fig. 1](#)

EFFICIENCY AI-assisted R&D would be reflected in compute efficiency gains. This is informative over and above capabilities progress because capabilities progress may simply reflect increased inputs.

- 2 Capability against training spend.** For each major model, capability as a function of total training expenditure, and the year-on-year efficiency gain at fixed capability. [Fig. 2](#)

- 3 Pre-training compute efficiency.** Loss against pre-training compute for both each production run and the small validation experiments that precede it. [Fig. 3](#)

- 4 Post-training compute efficiency.** Capability against post-training compute, measured on ECI and on unsaturated AI R&D benchmarks. [Fig. 3](#)

INPUTS TO AI RESEARCH Gains driven by more human labor are not evidence of RSI; gains driven by more inference compute are. Historical and ongoing series of AI R&D inputs help identify the source of AI progress.

- 5 Inference compute.** Internal usage in quantity (e.g. tokens) and in expenditure.
- 6 Experimental compute.** Internal usage in quantity (e.g. H100 hours) and in expenditure.
- 7 Headcount and compensation.** Employees and total compensation by team, function, and seniority.
- 8 How AI generates efficiency gains.** Surveys of researcher uplift, and a record of autonomous discoveries.
- 9 Research team budgets.** Budgets by team over time.

Frontier AI companies have indicated willingness to share more information about recursive self-improvement (RSI). This note describes nine data elements that would be highly informative about the pace of RSI, ordered by priority. The requests draw directly from the economic models of RSI in [Cunningham et al. \(2026\)](#).

Figure 1. Hypothetical model capabilities over time. Internal models leave the historical trend first.

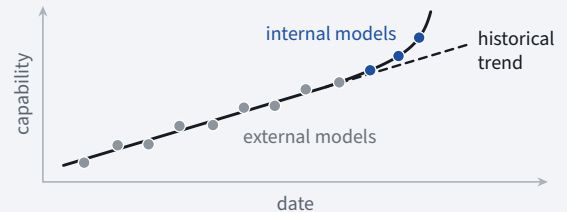


Figure 2. Hypothetical scaling curves of capability against training spend. Arrows show spend saved at equal capability.

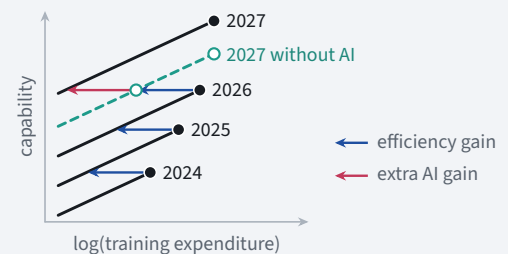
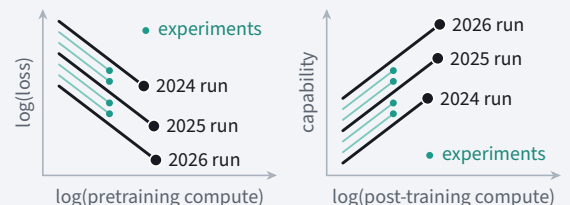


Figure 3. Hypothetical pre-training loss (left) and post-training capabilities (right) against compute. Smaller experiments (teal) precede each production run.



SCORECARD

AI	■ ■ ■ ■ ■ ■ ■ ■ ■ ■	1.5 / 9
🌀	■ ■ ■ ■ ■ ■ ■ ■ ■ ■	1.5 / 9
🌀	■ ■ ■ ■ ■ ■ ■ ■ ■ ■	0.5 / 9

We score company disclosures against the nine asks: 1 for a full release, 0.5 for a partial release, 0 for no qualifying data found. We score only the most recently released frontier models from Anthropic, OpenAI, and Google DeepMind as of September 20th 2026. Ask 1 examines unreleased models. See the linked scorecard for detailed explanations.

● released ○ partial ○ no credit